

SILAS M. FALDE

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SUMMARY

Highly motivated and detail-oriented Data Science junior and researcher at the University of Michigan, with a strong foundation in the programming languages C++ and Python. Proficient in data analysis and software development, with a passion for applying technical skills to a wide variety of problems. Currently, seeking opportunities to both apply and enhance my skills and knowledge to contribute to innovative projects and teams.

EDUCATION

University of Michigan School of Engineering
Senior B.E.S. in Data Science, GPA: 3.09/4.00

Ann Arbor, MI
August 2023 – December 2026

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

Western High School
GPA: 3.963

Parma, MI
August 2019 – May 2023

- Men's Varsity Soccer – Captain 1 year, Member 2 years
- Select Choir – Member 3 years

RESEARCH EXPERIENCE

University of Michigan Department of Epidemiology
Research Assistant – Computational Lead

Ann Arbor, MI
September 2025 – May 2026

- Led development of a Python pipeline to classify collective-violence narratives using prompt-based large language model inference and structured response processing.
- Built reusable preprocessing utilities for NVDRS-style text, demographic derivation, path management, and annotation-round data handling.
- Designed human annotation workflows and measured inter-annotator agreement with Cohen's kappa and Krippendorff's alpha to resolve disagreements and assess model performance.

University of Michigan Department of Epidemiology
Research Assistant

Ann Arbor, MI
September 2024 – May 2025

- Earned 3rd place in the CDC Youth Mental Health: Novel Variables Data Challenge, placing in roughly the top 1% of entries.
- Developed Python pipelines on a high-performance computing cluster to clean, normalize, and engineer features from more than 400,000 suicide-victim narratives.
- Used large language models to extract structured trends from narrative text and evaluated model-assisted outputs with classification metrics and error analysis.
- Documented findings with reproducible analyses, reports, and visualizations for the research team and publication.

WORK EXPERIENCE

Ally Financial, Inc.
Intern, Artificial Intelligence & Advanced Analytics

Detroit, MI
May 2026 – August 2026

- **Prototyped large language model workflows** to automate internal tasks and improve efficiency and safety.
- **Built Python NLP pipelines** to evaluate PII masking and redaction model performance for sensitive company documents.
- **Enhanced** a data dictionary migration tool by adding **batch processing**, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

Ally Financial, Inc.
Intern, Data Management

Detroit, MI
May 2025 – August 2025

- **Created** data dictionaries and lineage documentation in **Python** to improve data governance and compliance.
- **Added data quality checks** to existing data pipelines with **Python and SQL** to improve data integrity and reliability.

- **Developed a Power BI** data availability dashboard to monitor refresh rates and availability for critical data assets.

Michigan Soccer Referee Association
Certified Grassroots Soccer Referee

Ann Arbor, MI
March 2022 – Present

- **Officiating** on a referee team to create a safe, fair, and fun environment for players, coaches, and spectators.
- **Applying knowledge of the rules of soccer** to facilitate the sport.

Center Lake Fitness
Groundskeeper

Jackson, MI
May 2023 – August 2023

- **Mowed, trimmed, and landscaped** the grounds to maintain a clean and safe environment for patrons.
- **Conducted daily care** for farm animals including feeding, cleaning, monitoring animal health, and maintaining facilities.

PROJECTS

Home Server Infrastructure
Systems Developer

Ann Arbor, MI
2024 – Present

- **Designed and operate** a single-host **Ubuntu self-hosting platform** using **Docker Compose** stacks for ingress, media, cloud storage, AI services, monitoring, automation, and game hosting.
- **Automated media workflows** with **Bash and Python** for queueing, GPU-accelerated HEVC encoding, staging, metadata-safe publishing, library audits, and failure-aware batch processing.
- **Configured** Cloudflare Tunnel and Caddy ingress, Tailscale private administration, systemd scheduling, service health checks, and documented upgrade, backup, recovery, and security procedures.

silasfalde.net
Web Developer

Ann Arbor, MI
2024 – Present

- **Built and maintain a responsive static personal website** with modular HTML, CSS, and JavaScript for resume, research, project, education, and photography pages.
- **Developed browser-side navigation, responsive layouts, photography galleries, asset loading, and reusable site-content modules** without a Node build system.
- **Automated resume publishing** by parsing LaTeX section files into browser-friendly JSON, and generated photography event manifests from organized image folders.

PyNVDRS
Package Developer

Ann Arbor, MI
2025 – 2026

- **Refactored reusable utilities** from NVDRS-style epidemiology projects into a typed, installable Python package with editable-install support.
- **Organized shared modules** for paths, narrative text cleaning, demographic preprocessing, annotation workflows, and optional GPT-assisted inference.
- **Added import and path tests, package metadata, optional dependency groups, and continuous integration** to support migration from research scripts.

FPS PvP Dynamics
Project Developer

Ann Arbor, MI
March 2026 – April 2026

- **Built** a configurable **agent-based simulation** in Python to model player movement, combat, objectives, respawn-ing, and stochastic per-tick match dynamics.
- **Developed trace capture, metrics export, visualization, and automated tests** for debugging and vali-dating simulation behavior.
- **Ran parallel parameter sweeps and replication studies, Latin hypercube sampling, and Sobol sensi-tivity analysis** to quantify the effect of model parameters.

Photo Processing Helper
Project Developer

Ann Arbor, MI
Jan 2026 - Present

- **Built a modular Python CLI and library** for photo framing, collage generation, panorama stitching, and Nikon RAW-to-JPG matching across large image folders.
- **Implemented OpenCV, NumPy, Pillow, and rawpy pipelines** for resizing, cropping, alignment, blending, and exact-dimension output while preserving high-bit-depth detail.
- **Added tests and validation** for file discovery, output integrity, stitching behavior, and repeatable batch processing.

Predicting Survival in the ICU
Project Developer

Ann Arbor, MI
January 2025 – March 2025

- **Developed** logistic-regression, convolutional, and recurrent-neural-network models in **Scikit-learn** and **PyTorch** to predict ICU mortality from clinical time-series and static features.
- **Built reproducible data loaders and preprocessing** for missing values, temporal observations, and engineered clinical features, with configuration-driven training and evaluation.
- **Evaluated models** with **AUROC**, visualization, and automated tests covering CNN, transformer, and output behavior.

Capuchin Audio Classification

Machine Learning Developer

Ann Arbor, MI

Course Project

- **Built an end-to-end PyTorch pipeline** to detect capuchin monkey calls in forest recordings by loading, re-sampling, and batching variable-length audio.
- **Converted recordings to padded spectrogram tensors** and trained a convolutional neural network with GPU support, checkpointing, and TensorBoard loss tracking.
- **Ran inference** over field recordings and exported per-recording call counts to CSV for downstream analysis.

Vaccine Uptake Prediction

Data Science Developer

Ann Arbor, MI

Course Project

- **Built a reproducible Python pipeline** to predict multilabel vaccine uptake from NHFS survey data, including preprocessing, fold creation, and model evaluation.
- **Compared regularized logistic regression, random forests, and Bayesian hierarchical models** using cross-validation, ROC AUC, Brier scores, and calibration plots.
- **Organized configuration, data utilities, model utilities, and evaluation code** into a modular source package with a notebook-driven workflow.

Image Classification and Transfer Learning

Computer Vision Developer

Ann Arbor, MI

Course Project

- **Built a complete Python/PyTorch image-classification workflow** with dataset preparation, CNN and Vision Transformer models, training scripts, and challenge inference.
- **Compared custom CNNs, transfer-learning source and target models, and Vision Transformers**, with validation AUROC reported around **0.97** for CNN experiments.
- **Added automated tests** for datasets, image standardization, CNNs, ViTs, and generated outputs to validate the training and evaluation pipeline.

Sequence Learning and Clustering

Machine Learning Developer

Ann Arbor, MI

Course Project

- **Implemented and evaluated recurrent neural networks and self-attention models** for sequence prediction, achieving a reported test AUROC of **0.803**.
- **Implemented k-means and hierarchical clustering experiments**, comparing initialization strategies and objective values across runs.
- **Organized reproducible training configurations, checkpointed models, visualizations, and written analyses** across a Python source tree.

Probabilistic Models and Autoencoders

Machine Learning Developer

Ann Arbor, MI

Course Project

- **Implemented Gaussian mixture model training** with expectation-maximization and analyzed convergence and mixture assignments in Python.
- **Built experiments for Bayesian networks and autoencoders**, including denoising and regularized reconstruction models with saved checkpoints.
- **Used numerical modeling, visualization, and LaTeX-based technical reporting** to compare probabilistic and representation-learning methods.

Model-Based and Model-Free Reinforcement Learning

Reinforcement Learning Developer

Ann Arbor, MI

Course Project

- **Implemented Active ADP**, a model-based reinforcement-learning agent that learned transition and reward models and planned with value iteration.
- **Implemented Q-learning with optimistic exploration** and compared model-based and model-free learning in deterministic and stochastic Gymnasium FrozenLake environments.
- **Built evaluation, learning-curve, video-recording, and reproducibility utilities** with controlled random seeds and configurable experiments.

Database Storage and Grace Hash Join

C++ Developer

Ann Arbor, MI

Course Project

- **Implemented C++ database-storage components** for disk pages, records, memory buffers, buckets, and relation management.
- **Built the partition phase of a Grace Hash Join** using page-level I/O, hash-based bucket placement, buffer flushing, and disk-page tracking.
- **Worked across a multi-file C++ codebase** with Make-based builds and explicit memory, page-capacity, and record-layout constraints.

Photo Library Compressor

Python Developer

Ann Arbor, MI

Personal Project

- **Engineered a Python batch pipeline** to compress and reorganize large iCloud photo libraries using file size, format, and GPS metadata.
- **Preserved EXIF metadata** while applying configurable quality steps, routing missing-location images, and tracking processing results in CSV metrics.
- **Added timeout handling, staged processing, and failure-tolerant orchestration** for long-running HEIC and JPEG media workloads.

Breast Cancer Detection

Machine Learning Developer

Ann Arbor, MI

Course Project

- **Investigated breast-cancer image classification** with Python notebooks covering exploratory data analysis, convolutional neural networks, EfficientNetV2, and Vision Transformer models.
- **Built image preprocessing, training, and evaluation experiments** with NumPy, pandas, Pillow, Matplotlib, and PyTorch-oriented workflows.
- **Compared transfer-learning approaches** and documented model behavior through visual analysis and experiment reports.

Image Recognition and Metric Learning

Computer Vision Developer

Ann Arbor, MI

Course Project

- **Implemented image-classification and metric-learning experiments** in Python and PyTorch for a computer-vision recognition assignment.
- **Trained separate classification and embedding models**, saved learned weights, and analyzed recognition behavior through notebook experiments and reports.
- **Applied feature-space learning techniques** to compare image similarity and supervised recognition performance.

Computer Vision Panorama Stitching

Computer Vision Developer

Ann Arbor, MI

Course Project

- **Implemented a Python/OpenCV panorama pipeline** for numerical image processing, feature alignment, and composition of multiple photographs.
- **Used NumPy representations and geometric transformations** to align image content and produce stitched panoramic outputs.
- **Documented algorithmic choices and evaluated results** on supplied image datasets through a notebook-based workflow.

SKILLS

Programming: Python, C++, Bash, JavaScript, R, SQL

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, spaCy, NLTK

Web: HTML, CSS, JavaScript, responsive design, static-site publishing

Systems: Linux, Docker Compose, systemd, CMake, Make, Git, CI/CD

Software: VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta